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Autonomous forest fire detection, alerting and mitigation for communities

Abstract

Various embodiments for an autonomous forest fire detection, alerting and mitigation device includes a subterranean data gatherer having a plurality of soil temperature and soil moisture sensors installed at a plurality of different depths in a soil. A data assimilator that has a configured threshold change depending on the plurality of different depths in the soil of the plurality of soil temperature and soil moisture sensors, the soil properties, the duration of the fire, and a time delay duration. The data assimilator receives a data transmission packet from the subterranean data gatherer and detects a fire and generates a fire alert when the plurality of soil temperature and soil moisture sensors all breach the configured threshold change. A smart water pump is also started to slow the fire or completely stop it. The subterranean data gatherer is installed at the plurality of different depths in the soil.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) Forest fires contribute to 20% of carbon dioxide (CO2) in the atmosphere and cause irreparable ecological damage. Due to increased urbanization of the once-forested regions, more homes are at risk of wildfire damage. The fires are either detected very late and have grown significantly, or there is a lack of water and availability of firefighters for adequate response. As multiple fires start in an area, the fire response systems struggle due to the lack of resources. Existing fire detection methods for example human-based observation, satellite detection, optical cameras, and Wireless Sensor Networks (WSN) have low to medium reliability. Human-based observation methods have detection delays. Satellite detection is costly and detects fires once they become large and is impacted by clouds. Clouds and other environmental conditions impact optical camera-based solutions. WSN is plagued by false alarm repetitions. The issues with WSN-based solutions are either due to the type of sensors used or the technology used for data or image capture. Ambient sensors for example smoke, gas, thermal, and flame detectors raise false alarms due to fog, clouds, sunlight, and non-smoke objects. Solutions based on ambient surroundings monitoring have to deal with air pollution issues and other environmental conditions and are highly unreliable. These remotely installed systems also get destroyed or damaged in forest fires.

SUMMARY OF THE INVENTION

(2) Various embodiments provide a method and apparatus for an autonomous forest fire detection, alerting and mitigation for communities. Various aspects include installing a plurality of a

subterranean data gatherer around a community. The subterranean data gatherer utilizes a plurality of soil temperature and soil moisture sensors installed at a plurality of different depths in a soil. The subterranean data gatherer transmits data using a data transmission packet to a data assimilator. The community is surrounded by underground installation of a series of interconnected rainwater harvesting tanks that are connected to a plurality of a smart water pump. When a fire starts, the data assimilator detects fire and generates a fire alert using the data transmission packet sent by the subterranean data gatherer generates a fire alert and sends the data transmission packet to the smart water pump that is closest to the location of the fire to start a sprinkler. The sprinkler uses the series of interconnected rainwater harvesting tanks to mitigate the fire and to either slow the fire down or completely put the fire out. The data assimilator communicates with a data processing device to provide a real time alerts and data to a community members over a mobile app and desktop.
(3) In some aspects, the plurality of different depths in the soil of the plurality of soil temperature and soil moisture sensors is decided by the soil properties including thermal conductivity, thermal diffusivity, volumetric heat capacity, the soil type, the soil density and the soil porosity.
(4) In some aspects, the data assimilator has a configured threshold change for the plurality of soil temperature and soil moisture sensors. The configured threshold change is defined based on the plurality of different depths in the soil of the plurality of soil temperature and soil moisture sensors, a duration of time since a start of the fire, and a time delay duration. The data assimilator generates the fire alert when the plurality of soil temperature and soil moisture sensors have all individually breached the configured threshold change.
(5) In some aspects, the subterranean data gatherer is placed inside a subterranean data gatherer housing and the subterranean data gatherer housing is installed at the plurality of different depths in the soil. The subterranean data gatherer housing is made of fire-resistant materials and is waterproof. In some aspects the subterranean data gatherer housing is installed on top of a trolley platform installed at the plurality of different depths in the soil such that a handle of the trolley platform is above the surface of the soil to easily locate the subterranean data gatherer for any maintenance.
(6) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The novel features believed characteristics of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings wherein:
(2) FIG. **1** is an illustration of components of the autonomous forest fire detection, alerting and mitigation for communities in accordance with an illustrative embodiment.
(3) FIG. **2** is an illustration of components of the autonomous forest fire detection, alerting and mitigation for communities in accordance with an illustrative embodiment.
(4) FIG. **3** is an illustration of the subterranean data gatherer deployment in the soil in accordance with an illustrative embodiment.
(5) FIG. **4** is an illustration of the subterranean data gatherer in accordance with an illustrative embodiment.
(6) FIG. **5** is an illustration of a block diagram of the subterranean data gatherer in accordance with an illustrative embodiment.
(7) FIG. **6** is an illustration of the data assimilator in accordance with an illustrative embodiment.

(8) FIG. **7** is an illustration of a block diagram of the subterranean data gatherer, the data assimilator and the smart water pump in accordance with an illustrative embodiment.
(9) FIG. **8** is an illustration of the smart water pump in accordance with an illustrative embodiment.
(10) FIG. **9**A is an illustration of a model of decision making for generating the fire alert and autonomously triggering the smart water pump in accordance with an illustrative embodiment.
(11) FIG. **9**B is an illustration of a model of decision making for generating the fire alert and autonomously triggering the smart water pump in accordance with an illustrative embodiment.
(12) FIG. **9**C is an illustration of a model of decision making for generating the fire alert and autonomously triggering the smart water pump in accordance with an illustrative embodiment.
(13) FIG. **12** is an illustration of a block diagram illustrating architecture of the data transmission packet suitable for use with various embodiments.
(14) FIG. **13**A is an illustration of maintenance of the subterranean data gatherer in accordance with an illustrative embodiment.
(15) FIG. **13**B is an illustration of maintenance of the subterranean data gatherer in accordance with an illustrative embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(16) Various embodiments will be described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same parts. References made to particular examples and implementation are for illustrative purposes, and are not intended to limit the scope of the claims.
(17) The various illustrative embodiments provide a method and apparatus for the autonomous forest fire detection, alerting and mitigation for communities.
(18) With reference now to the figures and, in particular, with reference to FIG. **1**, an illustration of components of the autonomous forest fire detection, alerting and mitigation for communities is depicted in accordance with an illustrative embodiment. As depicted, the autonomous forest fire detection, alerting and mitigation for communities **100** includes the community **200**. The community **200** is surrounded by a forest **300**. In order to detect fire, the plurality of the subterranean data gatherer **400** are installed around the community **200**. The subterranean data gatherer **400** utilize the plurality of soil temperature and soil moisture sensors. The subterranean data gatherer **400** transmits data using the data transmission packet **470** (FIG. **12**) with a message identifier to the data assimilator **500**. The community **200** is surrounded by underground installation of the series of interconnected rainwater harvesting tanks **600** that are connected to the plurality of the smart water pump **1100**. When the fire **700** starts, the data assimilator **500** on detection of the fire **700** using the data transmission packet **470** (FIG. **12**) sent by the subterranean data gatherer **400** generates the fire alert **514** (FIG. **7**) and sends the data transmission packet **470** (FIG. **12**) to the smart water pump **1100** that is closest to the location of the fire **700** to start the sprinkler **800** which uses the series of interconnected rainwater harvesting tanks **600** to mitigate the fire **700** and to either slow the fire **700** down or completely put the fire **700** out. The data assimilator **500** communicates with the data processing device **1000** to provide the real time alerts and data to the community members over the mobile app and desktop.
(19) With reference to FIG. **2**, an illustration of components of the autonomous forest fire detection, alerting and mitigation for communities **100** is depicted in accordance with an illustrative embodiment when in this particular example, the community **200** is surrounded by the forest **300** partially on one side. As depicted, the plurality of the subterranean data gatherer **400** are installed around the community **200** covering the area that is at risk of the fire **700**. The subterranean data gatherer **400** transmits data using the data transmission packet **470** (FIG. **12**) to the data assimilator **500**. The community **200** is surrounded by underground installation of the series of interconnected rainwater harvesting tanks **600** that are connected to the plurality of the smart water pump **1100** covering the area that is at risk of fire **700**. When the fire **700** starts, the data assimilator **500** on detection of the fire **700** using the data transmission packet **470** (FIG. **12**) sent by the subterranean

data gatherer **400** generates the fire alert **514** (FIG. **7**) and sends the data transmission packet **470** (FIG. **12**) to the smart water pump **1100** that is closest to the location of the fire **700** to start the sprinkler **800** which uses the series of interconnected rainwater harvesting tanks **600** to mitigate the fire **700** and to either slow the fire **700** down or completely put the fire **700** out. The data assimilator **500** communicates with the data processing device **1000** to provide the real time alerts and data to the community members over the mobile app and desktop.

(20) FIG. **1** and FIG. **2** are presented as example configurations of components of the autonomous forest fire detection, alerting and mitigation for communities and are not meant to imply limitations to other configurations. One or more aspects of the embodiments represent a fully autonomous forest fire detection, alerting and mitigation for communities that is not reliant on any firefighting resources and can slow the fire providing valuable time for evacuations. The presence of autonomous forest fire detection, alerting and mitigation for communities across multiple communities allows for autonomous management of fires within communities allowing for evacuation and saving valuable property. One or more aspects of the embodiments represent the additional advantage of firefighting resources of the city not being overstretched as multiple fires start in a jurisdiction as well as solution for lack of water in these cases.

(21) FIG. **3** is an illustration of the subterranean data gatherer **400** deployment in the soil **401** in accordance with an illustrative embodiment. The subterranean data gatherer **400** consists of the subterranean data gatherer housing **402**. The subterranean data gatherer **400** comprises of the plurality of soil temperature and soil moisture sensors installed at the plurality of different depths in the soil **401**. The plurality of soil temperature and soil moisture sensors are the parts of the subterranean data gatherer **400** that are exposed outside the subterranean data gatherer housing **402** and are in direct contact with the soil **401**.

(22) In some embodiments the subterranean data gatherer **400** has two sensors a soil temperature sensor 1 **450** and a soil moisture sensor 1 **452** installed at a 1-inch depth in the soil **401**. In some embodiments the subterranean data gatherer **400** has four sensors the soil temperature sensor 1 **450** and the soil moisture sensor 1 **452** installed at the 1-inch depth in the soil **401** and a soil temperature sensor 2 **456** and a soil moisture sensor 2 **458** installed at a 3-inch depth in the soil **401**. In some embodiments the subterranean data gatherer **400** has the plurality of soil temperature and soil moisture sensors installed at the plurality of different depths for example 1-inch, 3-inch and 5-inch in the soil **401** respectively. These embodiments and the number of soil sensors and the type of soil sensors used and their installation depths are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(23) In contrast with ambient environment conditions, the below ground soil temperature and soil moisture change gradually during the course of the day. During precipitation, the soil moisture will increase but the soil temperature will not rise rapidly. The condition under which both the soil temperature and soil moisture rise rapidly is during a surface fire. Hence one or more aspects of the present embodiment that utilizes subterranean soil-based sensors for detecting fire removes the issue of false alarm repetitions that plague ambient sensor-based solutions and solutions that combine ambient sensors and soil-based sensors.

(24) In some embodiments the subterranean data gatherer **400** utilizes contact-based sensors that come in direct contact with the soil **401** providing accurate measurements. Common examples include thermocouples, resistance temperature detectors (RTDs) and thermistors. These sensors are inserted into the soil **401** to measure temperature at different depths. In some embodiments the subterranean data gatherer **400** utilizes capacitive soil moisture sensor. In some embodiments the subterranean data gatherer **400** utilizes TDR (Time Domain Reflectometry) or FDR (Frequency Domain Reflectometry) for soil moisture sensor. The type of sensors used will drive the cost of the subterranean data gatherer **400**.

(25) The subterranean data gatherer **400** are installed in immediate vicinity of the community **200** and also cover a certain area of the forest **300**. In some embodiments the subterranean data gatherer

400 can be deployed 5 miles into the forest **300**. In some embodiments the subterranean data gatherer **400** can be deployed 10 miles into the forest **300**. The area being covered should be considered when considering the type of sensors to be used. In some embodiments the subterranean data gatherer **400** can be made with different types of sensors depending on the upfront cost, longevity and maintenance of the plurality of soil temperature and soil moisture sensors.

(26) The subterranean data gatherer housing **402** itself is installed at the plurality of different depths in the soil **401**. In some embodiments a 1-foot pit is dug in the soil **401**. The subterranean data gatherer housing **402** is installed at the bottom of the 1-foot pit. In some embodiments the subterranean data gatherer housing **402** is installed further deep in the soil **401** for example at 2 feet or 3 feet depth.

(27) The depth at which the plurality of soil temperature and soil moisture sensors of the subterranean data gatherer **400** should be deployed is decided by the soil **401** properties including a thermal conductivity, a thermal diffusivity, a volumetric heat capacity, a type, a density and a porosity. High soil density means high thermal conductivity, so densely packed soil heats and cools faster than loosely packed soil. Soil water content or the fraction of pores filled with water is a major factor in heat conductivity. Thermal conductivity decreases when the fraction of soil pores filled with air increases, in medium-textured soils this relation is linear. Soil thermal conductivity means how fast heat moves through the soil, but it also changes as a consequence of heating. Thermal conductivity increases 3 to 5 times when soil is heated up to 90° C. Therefore, the subterranean data gatherer should be installed deeper for a sandy loam soil location as compared to a silty clay soil location. Also, if a location has more densely packed soil, then the subterranean data gatherer should be installed deeper. Hence the depth at which the plurality of soil temperature and soil moisture sensors are installed should be site-specific. The reason to install at a depth based on soil **401** is for the plurality of soil temperature and soil moisture sensors to survive the fire **700**.

(28) In some embodiments the subterranean data gatherer housing **402** is made out of fire-resistant materials for example cement, steel, gypsum, cast iron, stone, brick and mortar. In some embodiments the subterranean data gatherer housing **402** may also have fire-resistant insulation for example ceramic fiber insulation. In some embodiments the subterranean data gatherer housing **402** may incorporate additional elements such as vermiculite, perlite and fire-retardant sealants to enhance their protective capabilities. In some embodiments the subterranean data gatherer housing **402** may incorporate advanced composite materials that combine the strength of steel with the heat resistance of specialized polymers and ceramics.

(29) In some embodiments the subterranean data gatherer housing **402** is made waterproof by using a closed-cell foam gasket. When the subterranean data gatherer housing **402** is closed, pressure forms a durable barrier between the water outside and the interior of the subterranean data gatherer housing **402**. In some embodiments the subterranean data gatherer housing **402** is made waterproof by applying a covering of polyutherane. In some embodiments the subterranean data gatherer housing **402** is made waterproof by applying a cementitious coating of sand, organic and inorganic substances with silica-based materials. In some embodiments the subterranean data gatherer housing **402** is made waterproof by using either EPDM rubber, rubberized asphalt, thermoplastic or bituminous membrane. Off course, a combination of these different types of fire resistant and water proofing materials can be used.

(30) Since the subterranean data gatherer housing **402** is placed in the soil **401**, the subterranean data gatherer housing **402** is designed to survive the fire **700**. The fire surviving capability comes from the depth of the installation of the subterranean data gatherer housing **402**, the fire proofing material being used for the subterranean data gatherer housing **402** and due to the thermal conductivity of the soil **401** being low. Due the thermal conductivity of the soil **401** being low and the volumetric heat capacity of the soil **401** being high, the plurality of soil temperature and soil moisture sensors are also likely to survive the fire. Hence one or more aspects of the present disclosure is in contrast with existing solutions where most components are destroyed in the fire

and hence do not function at the time of the fire.
(31) In some embodiments during the installation of the subterranean data gatherer **400** a service laptop can be used to set a sender ID of the subterranean data gatherer **400**, a destination ID of the data assimilator **500** and a geolocation of the subterranean data gatherer **400**. The sender ID of the subterranean data gatherer **400**, the destination ID of the data assimilator **500** and the geolocation of the subterranean data gatherer **400** data is then sent in the data transmission packet **470** (FIG. **12**) by the subterranean data gatherer **400** to the data assimilator **500** along with a temperature and moisture sensor data for the plurality of soil temperature and soil moisture sensors. The sender ID and the geolocation would be unique for individual subterranean data gatherer **400**.
(32) These embodiments are presented as example configurations and are not intended to be exhaustive or limited to the embodiments in the form disclosed.
(33) FIG. **4** is an illustration of the subterranean data gatherer **400** in accordance with an illustrative embodiment. In this depicted example the subterranean data gatherer **400** is an example of one low-cost implementation for the subterranean data gatherer **400**. The subterranean data gatherer **400** in this example embodiment comprises of four sensors, the soil temperature sensor1 **450** and the soil temperature sensor 2 **456** with adapter modules for Arduino (DS18B20), capacitive soil moisture sensors the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** for Arduino, a microprocessor **462** (ATmega328P/CH340, Arduino nano board), a radio module **460** (RFM95 W 915 Mhz LoRa wireless receiver transmitter), a rechargeable battery **464**, a charging board **466**, and a solar panel **468**.
(34) In some embodiments the soil temperature sensor1 **450**, the soil temperature sensor 2 **456**, the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** on one end are attached to the microprocessor and the other end extends out of the subterranean data gatherer housing **402** (FIG. **3**) and is installed at the plurality of different depths in the soil **401** (FIG. **3**). In some embodiments the microprocessor **462**, the radio module **460**, the rechargeable battery **464**, the charging board **466** are secured inside the subterranean data gatherer housing **402** (FIG. **3**) while the solar panel **468** is installed above ground.
(35) In some embodiments the soil temperature sensor1 **450** and the soil temperature sensor 2 **456** capture a soil temperature at the plurality of different depths in the soil **401**. The soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** capture a soil moisture at the plurality of different depths in the soil **401**. In some embodiments the microprocessor **462** uses standard arduino libraries to process the data of the plurality of soil temperature and soil moisture sensors and create the data transmission packet **470** (FIG. **12**). The data transmission packet **470** (FIG. **12**) is then sent to the data assimilator **500** using the radio module **460**. The microprocessor **462** is powered by the rechargeable battery **464**.
(36) In some low-cost embodiments Raspberry Pi, NodeMCU, MSP430 Launch Pad, STM32 microprocessors or any other commercially available microprocessor can be used. In other embodiments different types of microprocessors for example application specific integrated circuit processors, reduced instruction set microprocessors, digital signal processors or any other commercially available microprocessor can be used. In some embodiments LTE-M and Narrowband-IoT (NB-IoT). Sigfox, Weightless SIG can be used instead of LoRa. The rechargeable battery **464** is known to last for 10 years or more and adding the solar panel **468** ensures the rechargeable battery **464** remains charged. In some embodiments other renewable methods to charge the rechargeable battery **464** can be used including wind power, hydroelectric power and geothermal energy. Of course, in some embodiments a combination of renewable energy sources can be used.
(37) FIG. **5** is an illustration of a block diagram of the subterranean data gatherer **400** in accordance with an illustrative embodiment. In some embodiments the plurality of soil temperature and soil moisture sensors are attached to the microprocessor **462**. In some embodiments the wires connecting the microprocessor **462** and the plurality of soil temperature and soil moisture sensors

are made waterproof by applying a covering of polyutherane. In some embodiments the wires are made waterproof by applying a cementitious coating of sand, organic and inorganic substances with silica-based materials. In some embodiments the wires are made waterproof by using either EPDM rubber, rubberized asphalt, thermoplastic or bituminous membrane. In some embodiments the wires are covered in fire-retardant sealants or fire-resistant insulation. Of course, a combination of these different types of fire resistant and water proofing materials can be used. The radio module **460** is attached to the microprocessor **462** and is used to send the data transmission packet **470** (FIG. **12**) to the data assimilator **500**. The rechargeable battery **464** is attached to the microprocessor **462** and the rechargeable battery **464** is charged using the solar panel **468** through the charging board **466**. In some embodiments the subterranean data gatherer housing **402** consists of the radio module **460**, the microprocessor **462**, the rechargeable battery **464**, and the charging board **466**. The solar panel **468** which is exposed above the soil **401** is the single sacrificial component of the subterranean data gatherer **400** in the event of a fire.

(38) The subterranean data gatherer **400** can be configured to send the data transmission packet **470** (FIG. **12**) to the data assimilator **500** at a time interval that can be configured. In some embodiments the data transmission packet **470** (FIG. **12**) can be sent at the time interval of 5 seconds. In other embodiments the data transmission packet **470** (FIG. **12**) can be sent at the time interval of 10 seconds, or 15 seconds, or 20 seconds etc.

(39) These embodiments are presented as example configurations and are not intended to be exhaustive or limited to the embodiments in the form disclosed.

(40) FIG. **6** is an illustration of the data assimilator **500** in accordance with an illustrative embodiment. In this depicted example embodiment, the data assimilator **500** is an example of one low-cost implementation for the data assimilator **500**. The data assimilator **500** comprises of a microprocessor **504** (Arduino Nano ESP32 IoT) and a radio module **502** (RFM95 W 915 Mhz LoRa wireless receiver transmitter). In some embodiments the data assimilator **500** is placed in a central location with access to a WIFI and a dedicated power source. The data assimilator **500** receives the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** over radio frequency using the radio module **502**. In some embodiments uninterruptable power supply UPS can be used to supply power to the data assimilator **500** so even if power is out the data assimilator **500** can function.

(41) In some low-cost embodiments Raspberry Pi, NodeMCU, MSP430 Launch Pad, STM32 microprocessors or any other commercially available microprocessor can be used for microprocessor **504**. In other embodiments different types of microprocessors for example application specific integrated circuit processors, reduced instruction set microprocessors, digital signal processors or any other commercially available microprocessor can be used. The data assimilator **500** communicates with the data processing device **1000** to provide the real time alerts and data to the community members over the mobile app and desktop.

(42) In some embodiments LTE-M and Narrowband-IoT (NB-IoT). Sigfox, Weightless SIG can be used instead of LoRa. In some embodiments the data assimilator **500** uses 5G cellular network, Zigbee, LoRaWAN, Light Fidelity to send data to the data processing device **1000**. In some embodiments the data processing device **1000** can be a server in a physical data center or an IoT cloud-based solution.

(43) The data processing device **1000** provides the real time alerts and data to the community members over the mobile app and desktop. In some embodiments this can include real time data from the plurality of soil temperature and soil moisture sensors from the subterranean data gatherer **400**. In some embodiments this can include a threshold check **512** (FIG. **7**), the fire alert **514** (FIG. **7**) and the status of the smart water pump **1100** (FIG. **7**). In some embodiment this can include data collected over time. In some embodiments when the fire **700** starts, the data assimilator **500** on detection of the fire **700** using the data transmission packet **470** (FIG. **12**) sent by the subterranean data gatherer **400** generates the fire alert **514** (FIG. **7**) and sends the data transmission packet **470**

(FIG. **12**) to the smart water pump **1100** that is closest to the location of the fire **700** to start the sprinkler **800** which uses the series of interconnected rainwater harvesting tanks **600** to mitigate the fire **700** and to either slow the fire **700** down or completely put the fire **700** out. The data assimilator **500** in parallel also sends the data to the data processing device **1000** to provide the real time alerts to the community members over the mobile app and desktop so the community members can evacuate if required.

(44) In some embodiments during the installation of the data assimilator **500** the service laptop is used to set the sender ID and the geolocation of the data assimilator **500**. The Sender ID and the geolocation would be unique for the data assimilator **500**.

(45) These embodiments are presented as example configurations and are not intended to be exhaustive or limited to the embodiments in the form disclosed.

(46) FIG. **7** is an illustration of a block diagram of the subterranean data gatherer **400**, the data assimilator **500** and the smart water pump **1100** in accordance with an illustrative embodiment. The individual subterranean data gatherer **400**, **400***a*,**400***b*,**400***c* send the data transmission packet **470** (FIG. **12**) to the data assimilator **500**. In some embodiments the subterranean data gatherer **400** sends the data transmission packet **470** (FIG. **12**) for the soil temperature sensor **450** and **456** and the soil moisture sensor **452** and **458** at the time interval that is configured.

(47) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) applies a data scrubbing **506**. The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) checks the destination ID on the data transmission packet **470** (FIG. **12**) and consumes the data transmission packet **470** (FIG. **12**) if the destination ID matches the data assimilator **500** sender ID. The data assimilator **500** would also ensure that it does not consume the data transmission packet **470** (FIG. **12**) with the message identifier that it has already processed earlier. In some embodiments the data assimilator **500** filters out the temperature and moisture sensor data that are significant outliers. In some embodiments machine learning techniques can be used to identify and remove the temperature and moisture sensor data that is outlier. The data scrubbing **506** is applied to the data transmission packets **470** (FIG. **12**) individually to the soil temperature sensor **450** and **456** sensor data and individually to the soil moisture sensor **452** and **458** sensor data.

(48) In some embodiments the data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) calculates a total running average **508**. In some embodiments the total running average **508** is calculated based on using a configurable period of time. In some embodiments the configurable period of time may be 4 hours. In some embodiments the configurable period of time may be 24 hours. The total running average **508** is calculated individually for the soil temperature sensors **450** and **456** and individually for the soil moisture sensors **452** and **458** as the data transmission packet **470** (FIG. **12**) is received by the data assimilator **500**. These embodiments are presented as example configurations and are not intended to be exhaustive or limited to the embodiments in the form disclosed.

(49) In some embodiments the data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) calculates a latest running average **510**. In some embodiments the latest running average **510** is calculated based on a configurable number of the temperature and moisture sensor data. In some embodiments the configurable number of the temperature and moisture sensor data may be the latest 10 temperature and moisture sensor data from the plurality of soil temperature sensor and soil moisture sensors. In some embodiments the configurable number of the temperature and moisture sensor data may be the latest 50 temperature and moisture sensor data from the plurality of soil temperature sensor and soil moisture sensors.

(50) In some embodiments the latest running average **510** is calculated based on the configurable period of time. In some embodiments the configurable period of time may be last 10 minutes of the temperature and moisture sensor data. In some embodiments the configurable period of time may be last 2 minutes of the temperature and moisture sensor data. The latest running average **510** is calculated individually for the soil temperature sensors **450** and **456** and individually for the soil

moisture sensors **452** and **458** as the data transmission packet **470** (FIG. **12**) is received by the data assimilator **500**. These embodiments are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(51) In the embodiment where the total running average **508** is calculated based on the configurable period of time of 4 hours, under normal conditions, the total running average **508** over 4 hours should see gradual change due to heating by the sun or natural cooling. Normal conditions are those conditions when there is no surface fire and any soil temperature change would be due to sun or weather-related changes. Under fire conditions, the latest running average **510** would deviate significantly from this last 4-hour total running average **508**.

(52) The configured threshold change is defined in the data assimilator **500** for the plurality of soil temperature and soil moisture sensors and the configured threshold change is defined based on the type of the sensor, the plurality of different depths in the soil of the plurality of soil temperature and soil moisture sensors, the duration of time since a start of the fire and the time delay duration. For each of the soil temperature sensors **450** and **456** and each of the soil moisture sensors **452** and **458** the data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) calculates a deviation as a percentage change of the latest running average **510** from the total running average **508** and compares the deviation to the configured threshold change as part of the threshold check **512**. If the deviation exceeds the configured threshold change for all the plurality of soil temperature and soil moisture sensors then the data assimilator **500** will detect fire and generate the fire alert **514** for the subterranean data gatherer **400**. The data assimilator **500** with send the fire alert **514** to the data processing device **1000** for the community members. The data assimilator **500** in parallel creates and sends the data transmission packet **470** (FIG. **12**) with a fire alert ON to trigger the smart water pump **1100** closest to the subterranean data gatherer **400** that is deviating from the configured threshold change as part of trigger smart water pump **516**.

(53) In some embodiments multiple subterranean data gatherers **400**, **400***a*, **400***b*, **400***c* communicate with a single data assimilator **500**. The density of the subterranean data gatherer **400** deployment is a function of how much area can be allowed to burn before a fire is detected by the subterranean data gatherer **400**. In some embodiments a cluster of subterranean data gatherers **400** can communicate with their own data assimilator **500** and multiple data assimilators **500** can communicate with the data processing device **1000**. In some embodiments a cluster of subterranean data gatherers **400** can communicate with their own data assimilator **500** and multiple data assimilators **500** can communicate with the central data assimilator **500** that communicates with the data processing device **1000**.

(54) In some embodiments the data assimilator **500** maintains a mapping of the smart water pump **1100** closest to the subterranean data gatherer **400**. When the subterranean data gatherer **400** that is deviating from the configured threshold change is identified by the data assimilator **500** then the smart water pump **1100** mapped closest will be started by sending the data transmission packet **470** (FIG. **12**) with the fire alert ON. For each of the soil temperature sensor **450** and **456** and each of the soil moisture sensor **452** and **458** the data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) calculates the deviation of the latest running average **510** from the total running average **508** and compares the deviation to the configured threshold change as part of the threshold check **512**. If the deviation does not exceed the configured threshold change then the data assimilator **500** will send the data transmission packet **470** (FIG. **12**) with the fire alert OFF to the smart water pump **1100** to stop the smart water pump **1100**. The smart water pump **1100** can be configured to stop when the data transmission packet **470** (FIG. **12**) with the fire alert OFF is received or the water in the rainwater harvesting tank **600** is finished.

(55) These embodiments are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(56) FIG. **8** is an illustration of the smart water pump **1100** in accordance with an illustrative embodiment. In this depicted example of an embodiment the smart water pump **1100** is an example

of one low-cost implementation for the smart water pump **1100**. The smart water pump **1100** comprises of the rainwater harvesting tank **600** connected to a DC freshwater pressure diaphragm pump **1112** which is connected to the sprinkler **800**. The DC fresh water pressure diaphragm pump **1112** is operated by a rechargeable battery **1114** attached to a charging board **1108**, and a solar panel **1110**. A microprocessor **1104** (ATmega328P/CH340, Arduino Nano board) is attached to a radio module **1102** (RFM95 W 915 Mhz, LoRa wireless receiver transmitter). The microprocessor **1104** is powered with a rechargeable battery **1106** with the charging board **1108**, and the solar panel **1110**.

(57) The transmission packet **470** (FIG. **12**) sent by the data assimilator **500** with the fire alert ON when received by the smart water pump **1100** through the radio module **1102** is processed by the microprocessor **1104** which in turn starts the DC freshwater pressure diaphragm pump **1112** through a 5V one-channel relay module **1116**.

(58) In some low-cost embodiments Raspberry Pi, NodeMCU, MSP430 Launch Pad, STM32 microprocessors or any other commercially available microprocessor can be used. In other embodiments different types of microprocessors for example application specific integrated circuit processors, reduced instruction set microprocessors, digital signal processors or any other commercially available microprocessor can be used. In some embodiments LTE-M and Narrowband-IoT (NB-IoT). Sigfox, Weightless SIG can be used instead of LoRa. In some embodiments any commercially available water pump can be used. In some other embodiments when the rainwater harvesting tank **600** is empty the water can be pulled from municipal water pipes into the rainwater harvesting tank **600**.

(59) In some embodiments during the install of the smart water pump **1100** the service laptop is used to set the destination ID and the geolocation of the smart water pump **1100**. The sender ID is also set to ensure the smart water pump **1100** only consumes the data transmission packet **470** (FIG. **12**) where the sender ID matches with what came in the data transmission packet **470** (FIG. **12**).

(60) These embodiments are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(61) FIG. **9**A is an illustration of a model of decision making for generating the fire alert **514** and autonomously trigger smart water pump **516** in accordance with an illustrative embodiment. In this depicted example embodiment, the subterranean data gatherer **400** has two soil temperature sensors installed. The soil temperature sensor 1 **450** and the soil temperature sensor 2 **456** are installed at the plurality of different depths in the soil **401** (FIG. **3**). In some embodiments the soil temperature sensor 1 **450** is installed at the 1-inch depth in the soil **401** (FIG. **3**) and the soil temperature sensor 2 **456** is installed at the 3-inch depth in the soil **401** (FIG. **3**). In some embodiments the soil temperature sensor 1 **450** is installed at a 2-inch depth and the soil temperature sensor 2 **456** is installed at a 4-inch depth. The depth at which the plurality of soil temperature and soil moisture sensors are installed can vary with different embodiments and what is presented is example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(62) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506**, calculate the total running average **508** for the soil temperature sensor 1 **450** and calculate the latest running average **510** for the soil temperature sensor 1 **450**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512** for the soil temperature sensor 1 **450**.

(63) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***b*, calculate the total running average **508***b* for the soil temperature sensor 2 **456** and calculate the latest running average **510***b* for the soil temperature sensor 2 **456**. The data assimilator **500** will then calculate the deviation and

compare the deviation to the configured threshold change as part of the threshold check **512***b* for the soil temperature sensor 2 **456**.
(64) If both the soil temperature sensor 1 **450** and the soil temperature sensor 2 **456** have all individually breached the configured threshold change, the data assimilator **500** will detect fire and send the fire alert **514** for the subterranean data gatherer **400** and trigger smart water pump **516**.
(65) In some embodiments the configured threshold change for the soil temperature sensor 1 **450** when installed at the 1-inch depth can be defined as 30%. The configured threshold change for the soil temperature sensor 1 **450** when installed at the 2-inch depth can be defined as 20%. In some embodiments the configured threshold change for the soil temperature sensor 2 **456** when installed at the 3-inch depth can be defined as 10%. The configured threshold change for the soil temperature sensor 2 **456** when installed at the 4-inch depth can be defined as 5%. This is because there is a temperature gradient in the soil **401** and the upper layers of the soil **401** will get heated before the heat penetrates the lower layers of the soil **401**. The configured threshold change is defined based on the depth of the plurality of soil temperature and soil moisture sensors and how quickly the fire should be detected while reducing false alarms.
(66) In some embodiments the configured threshold change for the soil temperature sensor 1 **450** when installed at the 1-inch depth can be defined as 30%. The configured threshold change for the soil temperature sensor 2 **456** when installed at the 3-inch depth can be defined as 10%. In this embodiment the configured threshold change for the plurality of soil temperature and soil moisture sensors, for the data assimilator **500**, is defined based on the time delay duration. In this embodiment the time delay duration is implemented by the data assimilator **500** by starting a timer for the time delay duration after both the soil temperature sensor 1 **450** and the soil temperature sensor 2 **456** have all individually breached the configured threshold change. If both the soil temperature sensor 1 **450** and the soil temperature sensor 2 **456** continue to all individually breach the configured threshold change once the time delay duration is over, the data assimilator **500** will detect fire and send the fire alert **514** for the subterranean data gatherer **400** and trigger smart water pump **516**. This delayed detection of the fire will increase the probability of detecting the fire without false alarms.
(67) In some embodiments the configured threshold change is defined based on the duration of time since the start of the fire. In these embodiments the configured threshold change for the soil temperature sensor 1 **450** when installed at the 1-inch depth can be defined as 10% and the configured threshold change for the soil temperature sensor 2 **456** when installed at the 3-inch depth can be defined as 30%. This is because if the fire has been ongoing for a while and is starting to slow down the upper layers of the soil **401** will have less increase in temperature while the lower layers of the soil **401** will show rapid temperature increases as the heat continues to penetrate the lower layers of the soil **401**. In other embodiments the configured threshold change for the soil temperature sensor 1 **450** when installed at the 1-inch depth can be defined as −10% and the configured threshold change for the soil temperature sensor 2 **456** when installed at the 3-inch depth can be defined as 10%. This is because as the fire dies down the upper layers of the soil **401** start to cool while the lower layers of the soil **401** will still show rising temperatures since it takes time for the heat reduction to penetrate the lower layers of the soil **401**. In these embodiments the fire is detected much later but with more certainty.
(68) These embodiments are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed. One or more aspects of the embodiment remove false alarm repetitions by using the configured threshold change of the plurality of soil temperature and soil moisture sensors deployed at a plurality of depth.
(69) FIG. **9**B is an illustration of a model of decision making for generating the fire alert **514** and autonomously trigger smart water pump **516** in accordance with an illustrative embodiment. In this depicted example embodiment, the subterranean data gatherer **400** has two soil temperature sensors and one soil moisture sensor installed. The soil temperature sensor 1 **450**, the soil temperature

sensor 2 **456** and the soil moisture sensor 1 **452** are installed at the plurality of different depth in the soil **401**. In some embodiments the soil temperature sensor 1 **450** and the soil moisture sensor 1 **452** are installed at the 1-inch depth in the soil **401** (FIG. **3**) and the soil temperature sensor 2 **456** is installed at the 3-inch depth in the soil **401** (FIG. **3**). In some embodiments the soil temperature sensor 1 **450** and the soil moisture sensor 1 **452** are installed at the 2-inch depth in the soil **401** (FIG. **3**) and the soil temperature sensor 2 **456** is installed at the 4-inch depth in the soil **401** (FIG. **3**). The depth at which the plurality of soil temperature and soil moisture sensors are installed can vary with different embodiments and what is presented is example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(70) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506**, calculate the total running average **508** for the soil temperature sensor 1 **450** and calculate the latest running average **510** for the soil temperature sensor 1 **450**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512** for the soil temperature sensor 1 **450**.

(71) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***b*, calculate the total running average **508***b* for the soil temperature sensor 2 **456** and calculate the latest running average **510***b* for the soil temperature sensor 2 **456**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512***b* for the soil temperature sensor 2 **456**.

(72) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***c*, calculate the total running average **508***c* for the soil moisture sensor 1 **452** and calculate the latest running average **510***c* for the soil moisture sensor 1 **452**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512***c* for the soil moisture sensor 1 **452**.

(73) If all three sensors the soil temperature sensor 1 **450**, the soil temperature sensor 2 **456** and the soil moisture sensor 1 **452** have all individually breached the configured threshold change, the data assimilator **500** will detect fire and send the fire alert **514** for the subterranean data gatherer **400** and trigger smart water pump **516**.

(74) In some embodiments the configured threshold change for the soil moisture sensor 1 **452** when installed at the 1-inch depth can be defined as 15%. In some embodiments the configured threshold change for the soil moisture sensor 1 **452** when installed at the 2-inch depth can be defined as 5%. This is because there is a moisture gradient in the soil **401** and the upper layers of the soil **401** (FIG. **3**) will have a concentration of moisture before the upper layers of the soil **401** (FIG. **3**) start to dry and moisture moves to the lower layers of the soil **401** (FIG. **3**). The configured threshold change is defined based on the depth of the plurality of soil temperature and soil moisture sensors and how quickly the fire should be detected while reducing false alarms.

(75) These embodiments are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.

(76) FIG. **9**C is an illustration of a model of decision making for generating the fire alert **514** and autonomously trigger smart water pump **516** in accordance with an illustrative embodiment. In this depicted example embodiment, the subterranean data gatherer **400** has two soil temperature sensors and two soil moisture sensors installed. The soil temperature sensor 1 **450**, the soil temperature sensor 2 **456**, the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** are installed at the plurality of different depths in the soil **401** (FIG. **3**). In some embodiments the soil temperature sensor 1 **450** and the soil moisture sensor 1 **452** are installed at the 1-inch depth in the soil **401** and the soil temperature sensor 2 **456** and the soil moisture sensor 2 **458** are installed at the 3-inch depth in the soil **401** (FIG. **3**). In some embodiments the soil temperature sensor 1 **450** and the soil

moisture sensor 1 **452** are installed at the 2-inch depth in the soil **401** (FIG. **3**) and the soil temperature sensor 2 **456** and the soil moisture sensor 2 **458** are installed at 4-inch depth in the soil **401** (FIG. **3**). The depth at which the plurality of soil temperature and soil moisture sensors are installed can vary with different embodiments and what is presented is example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed.
(77) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506**, calculate the total running average **508** for the soil temperature sensor 1 **450** and calculate the latest running average **510** for the soil temperature sensor 1 **450**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512** for the soil temperature sensor 1 **450**.
(78) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***b*, calculate the total running average **508***b* for the soil temperature sensor 2 **456** and calculate the latest running average **510***b* for the soil temperature sensor 2 **456**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512***b* for the soil temperature sensor 2 **456**.
(79) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***c*, calculate the total running average **508***c* for the soil moisture sensor 1 **452** and calculate the latest running average **510***c* for the soil moisture sensor 1 **452**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512***c* for the soil moisture sensor 1 **452**.
(80) The data assimilator **500** on receiving the data transmission packet **470** (FIG. **12**) from the subterranean data gatherer **400** will apply the data scrubbing **506***d*, calculate the total running average **508***d* for the soil moisture sensor 2 **458** and calculate the latest running average **510***d* for the soil moisture sensor 2 **458**. The data assimilator **500** will then calculate the deviation and compare the deviation to the configured threshold change as part of the threshold check **512***d* for the soil moisture sensor 2 **458**.
(81) If all four sensors the soil temperature sensor 1 **450**, the soil temperature sensor 2 **456**, the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** have all individually breached the configured threshold change, the data assimilator **500** will detect fire and send the fire alert **514** for the subterranean data gatherer **400** and trigger smart water pump **516**.
(82) In some embodiments the configured threshold change is defined based on the duration of time since the start of the fire. In these embodiments the configured threshold change for the soil moisture sensor 1 **452** at the 1-inch depth can be defined as 5% and the configured threshold change for the soil moisture sensor 2 **458** at the 3-inch depth can be defined as 10%. This is because if the fire has been ongoing for a while and is starting to slow down then the upper layers of the soil **401** (FIG. **3**) will start to dry while the lower layers of the soil **401** (FIG. **3**) will show rapid moisture increases as the heat continues to penetrate the lower layers of the soil **401** (FIG. **3**). In other embodiments the configured threshold change for the soil moisture sensor 1 **452** at the 1-inch depth can be defined as −10% and the configured threshold change for the soil moisture sensor 1 **452** at the 3-inch depth can be defined as 5%. This is because as the fire dies down the upper layers of the soil **401** (FIG. **3**) are drying rapidly while the lower layers of the soil **401** (FIG. **3**) will still show rising temperatures and rising moisture. In these embodiments the fire is detected much later but with more certainty.
(83) In some embodiments the configured threshold change for the soil moisture sensor 1 **452** when installed at the 1-inch depth can be defined as 30%. The configured threshold change for the soil moisture sensor 2 **458** when installed at the 3-inch depth can be defined as 10%. In this embodiment the configured threshold change for the plurality of soil temperature and soil moisture

sensors, for the data assimilator **500**, is defined based on the time delay duration. In this embodiment the time delay duration is implemented by the data assimilator **500** by starting a timer for the time delay duration after both the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** have all individually breached the configured threshold change. If both the soil moisture sensor 1 **452** and the soil moisture sensor 2 **458** continue to all individually breach the configured threshold change once the time delay duration is over, the data assimilator **500** will detect fire and send the fire alert **514** for the subterranean data gatherer **400** and trigger smart water pump **516**. This delayed detection of the fire will increase the probability of detecting the fire without false alarms.

(84) In some embodiments the subterranean data gatherer **400** has the plurality of soil temperature sensors and soil moisture sensor installed at a plurality of the different depths in the soil **401**. In this embodiment the data assimilator **500** will determine that all the plurality of soil temperature sensors and the soil moisture sensors have breached the configured threshold change before triggering the fire alert **514**.

(85) Some or more aspects of the example embodiments presented here detect fire irrespective of weather, season, topography, historical weather patterns.

(86) FIG. **12** is an illustration of a block diagram illustrating an architecture of the data transmission packet suitable for use with various embodiments. The data transmission packet **470** sent by the subterranean data gatherer **400** and the data assimilator **500** includes among other things the destination ID, the sender ID, the message identifier, a data type, data lengths for individual sensor data, soil temperature sensor 1 value, soil temperature sensor 2 value, soil moisture sensor 1 value, soil moisture sensor 2 value, location coordinates, the fire alert. These data attributes are presented as example configurations and is not intended to be exhaustive or limited to the embodiments in the form disclosed. In some embodiments when the data type is 0 the data transmission packet **470** is coming with data from the subterranean data gatherer **400**, when the data type is 1 the transmission packet **470** is coming from the data assimilator **500** for the smart water pump **1100**, when the data type is 2 it means the subterranean data gatherer **400** is malfunctioning and requires maintenance. The data assimilator **500** on receiving the transmission packet **470** with the data type as 2 acommunicates with the data processing device **1000** to provide the real time alerts and data to the community members over the mobile app and desktop allowing the community members to do maintenance of the subterranean data gatherer **400** and replace defective parts.

(87) FIG. **13**A is an illustration of maintenance of the subterranean data gatherer **400** in accordance with an illustrative embodiment. In some embodiments for the ease of maintenance the rechargeable battery **464** can be placed on a support **474** that is attached to a tree **472** nearby and attached to the solar panel **468**. The solar panel **468** is also attached to the tree **472**. In some embodiments if trees are not available then ground poles can be installed to mount the rechargeable battery **464** using a support **474** and attaching the rechargeable battery **464** to the solar panel **468** by mounting the solar panel **468** to the ground pole. This facilitates changing the rechargeable battery **464** easily if needed. In some embodiments the subterranean data gatherer housing **402** can be installed on top of a trolley platform **476** installed at the plurality of different depths in the soil **401** such that a handle of the trolley platform is above the surface of the soil **401**. This helps locate the subterranean data gatherer and allows pulling the subterranean data gatherer housing **402** out for any maintenance. In some embodiments the subterranean data gatherer **400** sends an alert when the rechargeable battery **464** capacity is low and needs to be replaced. In the embodiments where the rechargeable battery **464** is installed inside the subterranean data gatherer housing **402** that is buried in the soil **401**, the subterranean data gatherer housing **402** can be retrieved by pulling the trolley platform **476**. The rechargeable battery **464** can easily be replaced by opening a latch **478** on the top of the subterranean data gatherer housing **402**. When the data assimilator **500** receives the data transmission packet **470** (FIG. **12**) with data type as 2 indicating the subterranean data gatherer

400 is malfunctioning and requires maintenance the data assimilator **500** communicates with the data processing device **1000** to provide the real time alerts and data to the community members over the mobile app and desktop allowing the community members to do maintenance of the subterranean data gatherer **400** and the defective component can be replaced by opening the latch **478**. In some embodiments the latch **478** is a compression latch that features a gasket that is pushed against the opening to create a secure seal to keep dust and moisture out. In some embodiments the latch **478** features a T-handle for the community members to fold up the T-handle rotate it quarter turn and pull the door open. This will seal out dust and water from the subterranean data gatherer housing **402**.
(88) FIG. **13**B is an illustration of maintenance of the subterranean data gatherer **400** in accordance with an illustrative embodiment. In some embodiments one side of the handle of the trolley platform **476** can extend into a pole structure **473**. For the ease of maintenance, the rechargeable battery **464** can be placed on the support **474** attached to the pole structure **473** and attached to the solar panel **468** that is also attached to the pole structure **473**.
(89) In conclusion, the disclosure describes the autonomous forest fire detection, alerting and mitigation for communities **100** that has the plurality of the subterranean data gatherer **400** that are installed around the community **200**. The subterranean data gatherer **400** utilize the plurality of soil temperature and soil moisture sensors deployed at the plurality of different depths in the soil **401**. The subterranean data gatherer **400** sends the data transmission packet **470** (FIG. **12**) to the data assimilator **500**. The data assimilator **500** determines if there is fire based on all the plurality of soil temperature and soil moisture sensors of the subterranean data gatherer **400** breaching the configured threshold change and issues the fire alert **514** and triggers the smart water pump **1100** to start the sprinkler **800** to slow or completely put out the fire. One or more aspects of the present disclosure that utilizes subterranean soil-based sensors for detecting fire removes the issue of false alarm repetitions that plague ambient sensor-based solutions. One or more aspects of the present disclosure is in contrast with existing solutions where most components are destroyed in the fire and hence do not function at the time of fire. One or more aspects of the present disclosure remove false alarm repetitions by using the configured threshold change of the plurality of soil temperature and soil moisture sensors deployed at the plurality of different depths in the soil **401**. One or more aspects of the example embodiments presented here detect fire irrespective of the environmental conditions, the weather, the season, topography, historical weather data while the effectiveness of the existing solutions is dependent on season and topography and environmental conditions.
(90) The description of the different illustrative embodiments has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skills in the art.
(91) Further, different illustrative embodiments may provide different features as compared to other illustrative embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principals of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A device for forest fire detection comprising: a subterranean data gatherer having a plurality of soil temperature and soil moisture sensors installed at a plurality of different depths in a soil; and a data assimilator having a configured threshold change for said plurality of soil temperature and soil moisture sensors, said configured threshold change is defined based on said plurality of different depths in said soil of said plurality of soil temperature and soil moisture sensors, and said data assimilator receives a data transmission packet from said subterranean data gatherer and said data

assimilator detects fire and generates a fire alert when said plurality of soil temperature and soil moisture sensors have all individually breached said configured threshold change; and said plurality of different depths in said soil of said plurality of soil temperature and soil moisture sensors is decided by said soil properties including thermal conductivity, thermal diffusivity, volumetric heat capacity, said soil type, said soil density and said soil porosity.

2. The device of claim 1, wherein said configured threshold change for said plurality of soil temperature and soil moisture sensors, for said data assimilator, is defined based on a duration of time since a start of a fire.

3. The device of claim 1, wherein said configured threshold change for said plurality of soil temperature and soil moisture sensors, for said data assimilator, is defined based on a time delay duration, wherein said fire alert will be raised when said plurality of soil temperature and soil moisture sensors all individually continue to breach said configured threshold change once said time delay duration is over.

4. The device of claim 1, wherein said subterranean data gatherer is placed inside a subterranean data gatherer housing and said subterranean data gatherer housing is installed at said plurality of different depths in said soil and said subterranean data gatherer housing is made of fire-resistant materials and is waterproof.

5. The device of claim 1, wherein said subterranean data gatherer is placed inside a subterranean data gatherer housing and said subterranean data gatherer housing is installed on top of a trolley platform installed at said plurality of different depths in said soil such that a handle of said trolley platform is above the surface of said soil to easily locate said subterranean data gatherer for any maintenance.

6. The device of claim 1, wherein said subterranean data gatherer is placed inside a subterranean data gatherer housing wherein said subterranean data gatherer housing consists of a latch to perform maintenance.

7. The device of claim 1, wherein said plurality of different depths in said soil of said plurality of soil temperature and soil moisture sensors is decided by said soil properties including thermal conductivity, thermal diffusivity, volumetric heat capacity, said soil type, said soil density and said soil porosity, and said configured threshold change for said plurality of soil temperature and soil moisture sensors, for said data assimilator, is defined based on said plurality of different depths in said soil of said plurality of soil temperature and soil moisture sensors.